Poster cue card

Opener (30 s). Most AI safety is one number at the output. We keep the structure: who's affected, what's owed, who consented, who carries the risk. We collapse to a decision only at the end, and record how. Every decision is sealed in a hash-chained proof you can replay. Open source, on PyPI, running today.

Then ask: "Model side or agent side?" Model → **6**. Agent → **5** then **7**. **10 s version:** a safety layer that seals every decision in a replayable proof, and compares what a model says with what its activations show.

Walkthrough (point at the number)

1 Thesis Structure before the scalar. Log the weights, who lost, the residue. Not "knows what's good", but every collapse is replayable.

2 Compiler Text → MoralGraph → tensor → DecisionProof. 12 passes, hashed at every stage. Rules, LLM + critic, LaBSE probe.

3 Pluralism Four lenses: consequences, duties (Z3), virtue, care. Never averaged. When they disagree, defer to a human.

4 Geometry Swap and negation commute → Klein four-group V_4, proved in Lean. D_4 was our overclaim, now corrected. The quarter-turn was not found in an LM. **Bond Index:** how often the verdict changes under the agent/patient swap. Human baseline **0.155** (≈20k letters). Warns at 0.25, blocks at 0.30.

5 Gateway Planner → Reflex / Tactical (10–100 ms) / Strategic → allow, revise, block. SHA-256 chained proofs. Service down → rules. **Fails safe, never open.**

6 PyTorch lens Forward hooks: activation lens vs. text lens → requires-human-review. Mid-layer, read in the consumer metric: **+19–21 pts** vs. raw L2 at a matched flag rate (Qwen2.5-0.5B). Replicated on TinyLlama and on French backtranslation. The final layer fails, so probe mid-layers. Early.

7 Left Margaret collapsed. Normal mode: forbid. Emergency: prefer ×4, and the ranking flips from stand-by to full response. Elevation = privilege escalation: confidence **0.97** vs. 0.90, scoped to the paramedics, reversible, audited. At **0.89** it's refused: EMS is still called, the records are not shared. A false alarm unlocks nothing, and aid is never blocked.

7 Right In the robot, DEME chooses nothing. Governor → constitution → DEME → manifest; any stop means hold. Change one input, prediction written first. **9 pairs**: a forged detector stopped the fire call, and her refusal decided the ambulance. Caught a lift bug: fixed, and only the lift's verdict changed.

Close Keep the structure · defer on conflict · fail safe. QR codes → paper + code. Hand them a brochure.

Questions

Whose ethics? It isn't claimed to be right. Four lenses, no averaging, a human decides conflicts. The claim is auditability.

Real robot? No. Simulated 3D home, scripted proposals, no hardware. The agent stack and the two NPCs are live.

Speed? Tactical 10–100 ms, reflex faster.

Service crashes? Rule fallback. Fails closed.

Lens works? Early. Mid-layer result replicated 2 of 2. Final layer fails validation. Not a finished detector.

Why group theory? It's a symmetry test you can run. We claim only V_4. Corrected D_4 ourselves. Earlier gauge model falsified by CHSH tests on human judgments (N=600, all $|S| \leq 2$).

COMPAS? LBI **1.049** [1.027, 1.072], $z = 4.9$. ≈1.00–1.02 under matching parity. A screening signal, not a regulatory audit.

Install? `pip install erisml-compiler` (alpha 0.9.0, MIT). erisml-lib 3.1.0 on PyPI. The worked example needs the source tree.

Not done? Ranks 1–3 real, higher partial, action axis a stub. Hohfeld/Bond Index in erisml-lib, not the compiler yet. D_4 posited.

PyTorch engineer extra: turboquant-pro on PyPI. One-line HF KV cache drop-in, Triton kernels, vLLM plugin, ≈5× less KV memory.

Panel map

1 Thesis	3 Pluralism	5 Gateway
2 Compiler	4 Geometry	6 PyTorch lens
7 Care-robot example | robot firewall		
takeaways · limits · QR codes		